

ATLAS — Month 1 Sprint Plan

7-Person Team | Goal: De-risk the real bottleneck, ship two working pieces, prototype the third

The One Decision This Plan Is Built Around

Your own docs are explicit: **exact Vietoris-Rips persistence scales combinatorially**, and nobody has proven the mitigations (sparsification, dimension-capping, incremental updates) hold at production scale. That single unresolved question determines whether Manifold-Core — and everything built on top of it — is viable.

So Month 1 puts the most people on **answering that question with real data**, not on spreading thin across three pillars and having none of them land.

Team Allocation (7 people)

Workstream	Headcount	Why this many
 A — Data Pipeline & ANN Graph (Phase 1, low risk)	2	Foundational, well-understood, needs to be <i>solid</i> fast so others aren't blocked
 B — VR Persistence / TDA (the real risk)	3	This is the make-or-break question — it gets the most senior attention
 C — S-MCP Sandbox (real engineering, not research)	1-2	Independent of A/B's outcome, genuinely completable in 4 weeks
 D — TIP Math Prototype (synthetic only)	1 (can float from B)	Depends on B's output existing first — spike only, not networked

If you have a designated tech lead, put them across A+B as a floater — those two workstreams have the most sequencing dependencies.

Week-by-Week Plan

Week 1 — Foundations + First Real Signal

Workstream	Deliverable
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A	Corpus acquired (named source, 10k-50k docs), preprocessing + dedup, embeddings generated and stored in Parquet with a data-quality report
B	GUDHI/Ripser selected (not hand-rolled). Synthetic ground-truth validation first: circle → exactly 1 persistent loop, two clusters → exactly 2 components. This proves the pipeline is <i>correct</i> before you ever ask if it's <i>fast</i>
C	Wasmtime + WASI sandbox skeleton stood up. A trivial "hello world" module round-trips through it with memory/compute caps enforced
D	Re-derive and sanity-check the persistence-landscape + LSH math on paper/notebook against toy vectors — no networking yet

 **Week 1 gate:** if B can't get synthetic ground truth right, stop and fix that before touching real data — a fast wrong answer is worse than a slow right one.

Week 2 — Real Data, Small Scale

Workstream	Deliverable
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A	Approximate k-NN graph built on full corpus (hnswlib/faiss-HNSW), recall@k measured against a brute-force baseline on a 1-2k subsample — report the actual recall number , not a vibe
B	Exact VR persistence run on the <i>real</i> embeddings, but only 1k-2k points. Timing and memory recorded honestly. This is the first real (non-synthetic) data point on the bottleneck
C	First real distilled module: one narrow task (e.g. entity-linkage on a toy dataset), compiled to wasm32-wasi, int8-quantized, signed, executed inside the sandbox
D	Prototype the persistence-landscape transform on B's Week 2 output (still single-machine, no cross-org exchange)

 **Week 2 gate:** B's timing number at 1-2k points is the first real data toward "does this scale." If it's already unreasonably slow at this size, that's critical information — surface it immediately, don't bury it hoping Week 3 fixes it.

Week 3 — Push the Bottleneck, Prove the Sandbox

Workstream	Deliverable
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A	Multi-hop query traversal prototype: query dropped into the graph, potential field defined from density + persistence weights, walked like gradient descent
B	The core experiment: attempt sparsified/witness-complex VR persistence at increasing scale — 5k → 10k → 25k points. Plot time/memory vs. scale. This curve is the actual answer to the open question
C	Measure distillation accuracy retention vs. the teacher model on the Week 2 task. Target: know the real number, not a hoped-for one
D	Run the LSH collision-rate math against B's real (not synthetic) persistence landscapes — still offline, no network exchange yet

 **Week 3 gate:** whatever B's curve shows — even if it's bad news — that is the single most valuable output of the month. A plan that discovers "sparsification breaks down past 15k points" in week 3 is worth more than a plan that hides that until month 3.

Week 4 — Consolidate, Benchmark, Report Honestly

Workstream	Deliverable
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A + B	Full retrieval benchmark: ATLAS traversal vs. plain cosine-RAG on a named external dataset (HotpotQA or MuSiQue subsample) — real exact-match numbers, both ways
C	Package a public/internal zero-exfiltration demo: show the sandbox's memory ceiling, compute-step budget, and no filesystem/network access holding under adversarial test inputs
D	Write up the TIP math prototype status: does the LSH collision rate behave as predicted on real landscapes? What's the measured false-negative rate?
All	One honest status report: what's measured, what's still open, and a real recommendation on Phase 2/3 timing based on B's actual scaling curve — not the original 8-month guess

What "Done" Looks Like on Day 30

- A working Manifold-Core pipeline, benchmarked with real recall/latency/accuracy numbers (even if scale is smaller than the original 10k-50k target — report what it

actually handled)

- A working S-MCP sandbox with a measured distillation-accuracy number and a demonstrated zero-exfiltration result
- A TIP math prototype validated against real (not synthetic) persistence data, on a single machine — explicitly **not** yet a live cross-org handshake
- A real, data-backed answer to "does sparsified VR persistence hold at scale" — this is the single deliverable that should drive whatever you promise for Month 2

✗ What This Plan Deliberately Does Not Promise

- A live two-org TIP handshake over the network — that depends on B's output existing and being trustworthy first
- Production-scale (50k+) VR persistence, if Week 3's curve shows it doesn't hold — in which case Month 2 is about fixing *that*, not building on top of an unproven foundation
- Any accuracy or leakage-bound numbers you haven't personally seen a benchmark script produce

This plan optimizes for one thing: at the end of the month, every number in your next status update is real. That's worth more to the project's actual trajectory than a fragile end-to-end demo that looks complete but isn't load-bearing.